



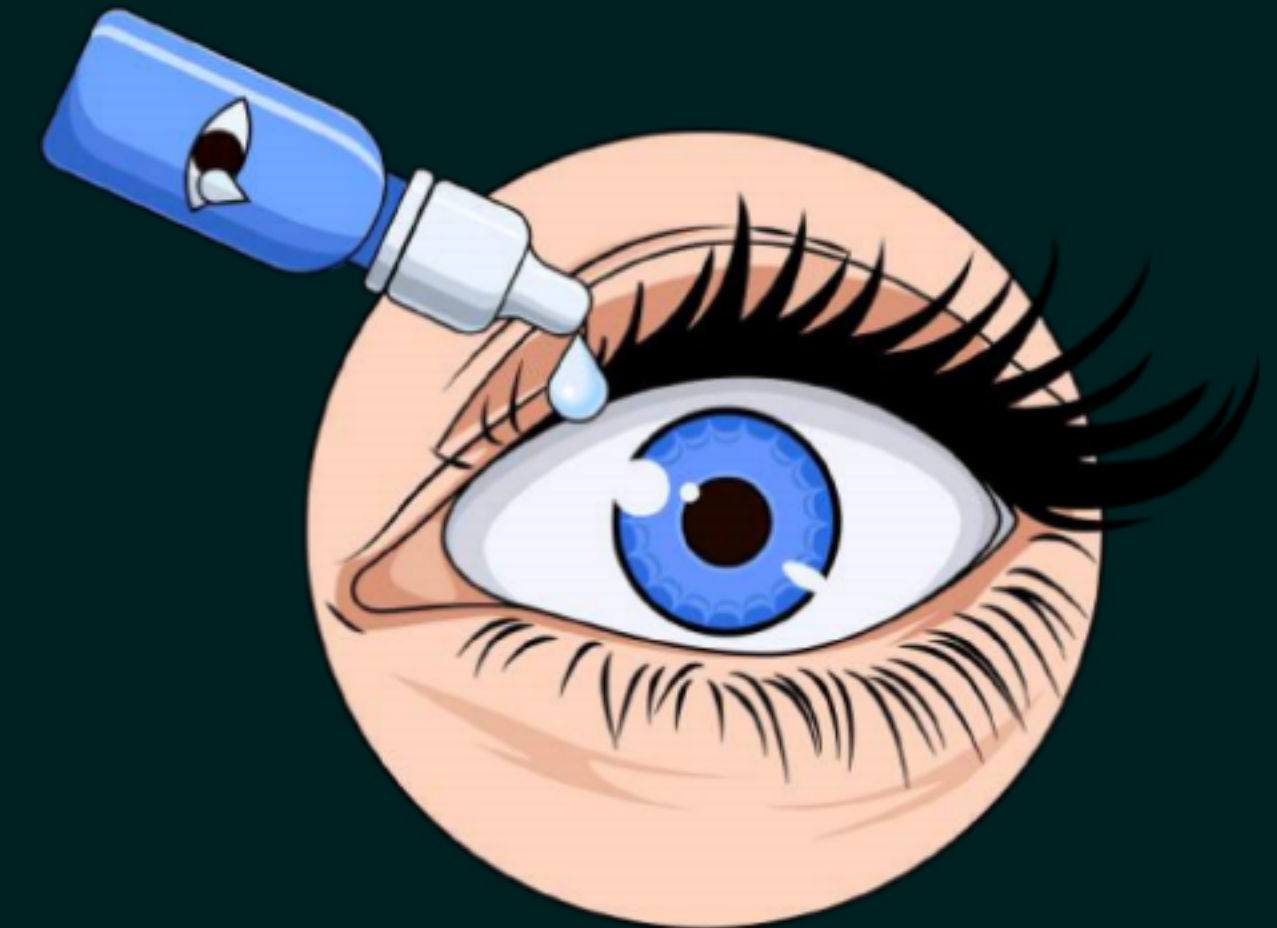
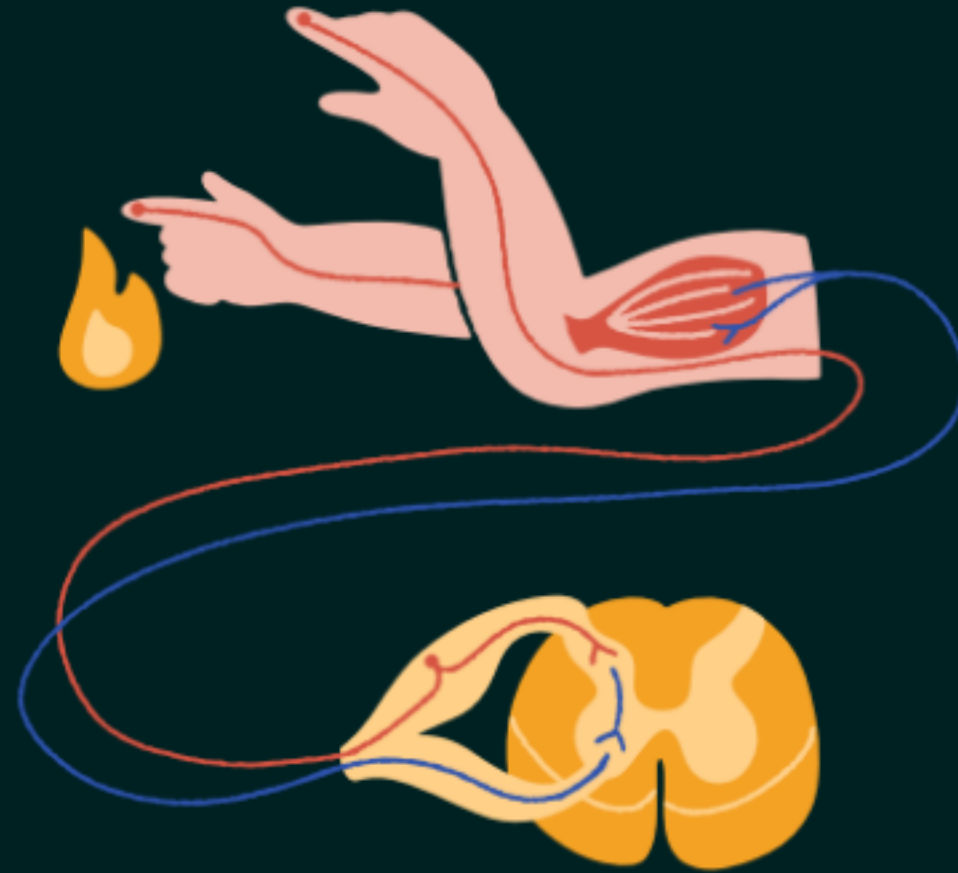
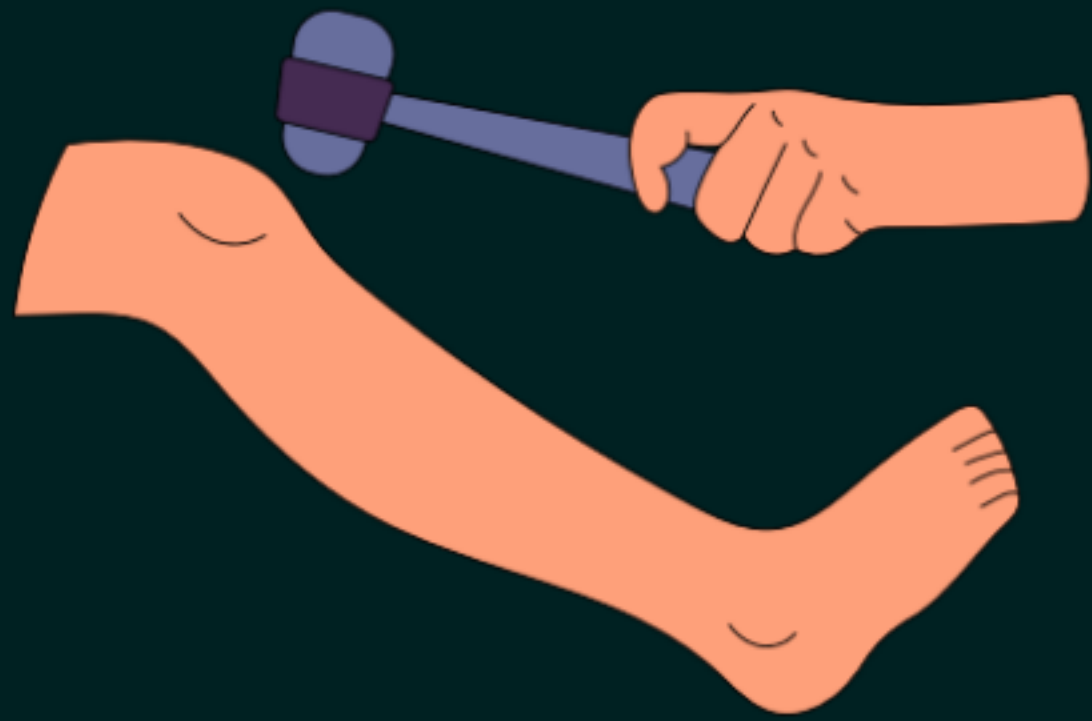
L1

CLASS X - SCIENCE

CONTROL AND COORDINATION

| PRASHANT KIRAD |

Why you react Automatically?



Coordination:

✓ The working together of different parts of the body in a proper and organised way to produce a suitable response to the environment.

✓ Example:

- Blinking when dust enters the eye
- ✓ • Sneezing due to irritation in the nose
- ✓ • Knee-jerk reaction during a doctor checkup
- ✓ • Closing eyes in bright light



// Control & coordination ensures:

- Correct movement for each specific change (stimulus).
- No wrong or unnecessary actions.

CONTROL & COORDINATION

Coordination in Animals

- Nervous System (Associated with Neurons)
- Endocrine System (Associated with Hormones)

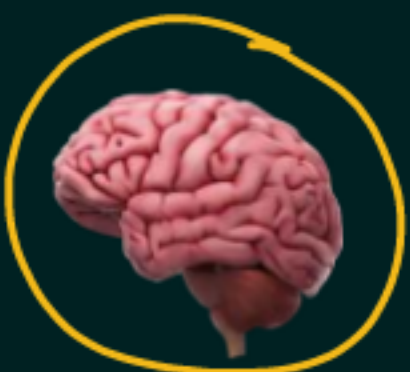
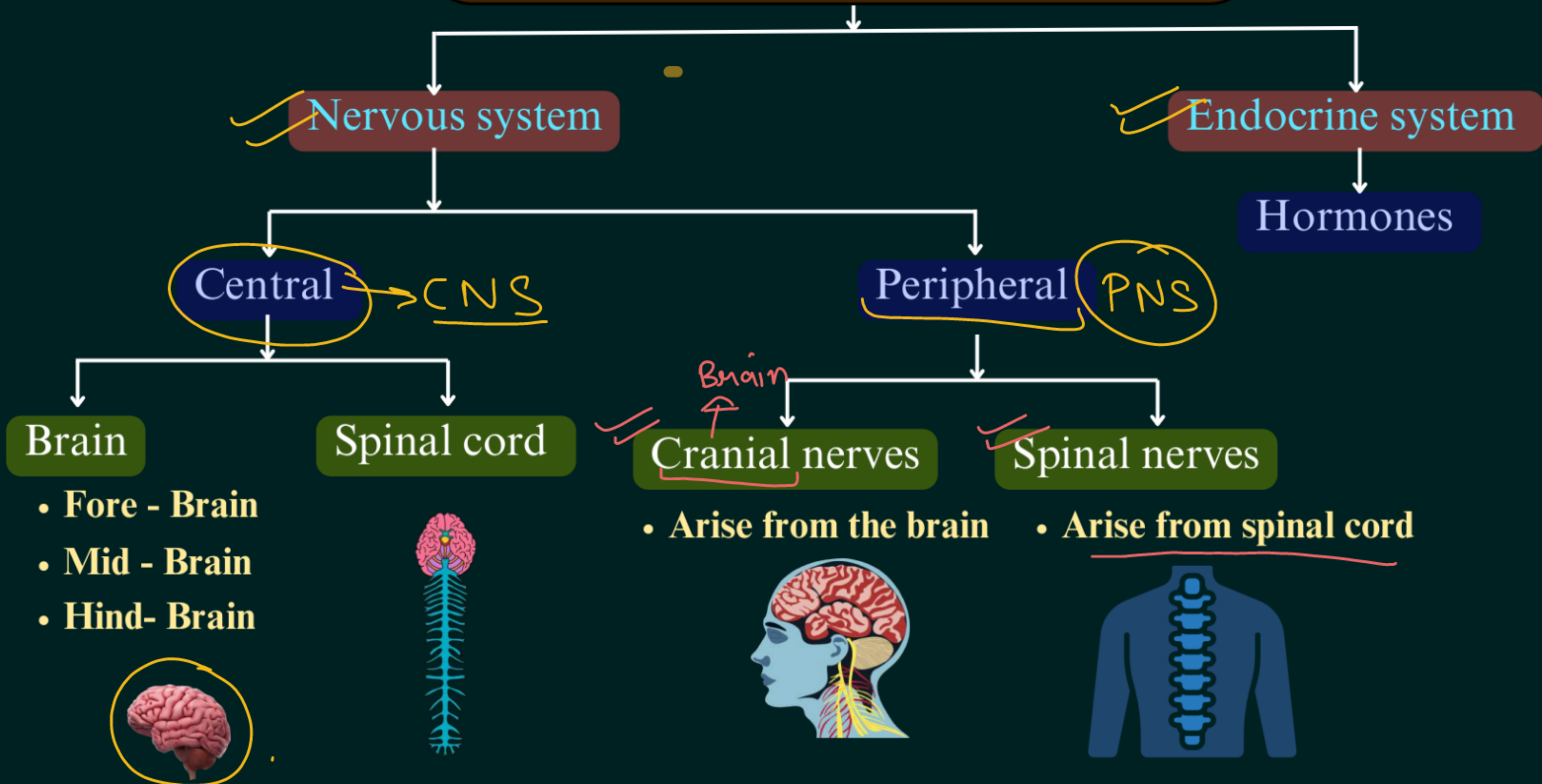


Coordination in Plants

- Plant Movements
- Plant Hormones



Coordination in Animals:





Brain

Spine

Central Nervous System (CNS)

Peripheral Nervous System (PNS)

→ Consists of the brain and spinal cord

Consists of nerves arising from the brain and spinal cord

→ Acts as the control and coordination centre

Connects the CNS to different body parts

→ Processes and interprets information

Carries impulses to and from the CNS

→ Protected by skull and vertebral column

✓ Spread throughout the body

Includes brain and spinal cord only

Includes cranial, spinal, and visceral nerves

Stimulus / stimuli

Stimulus is any change in the environment (internal or external) that produces a response in an organism. Example: Light, sound, smell, touch, temperature.

Example:



Heat



Sound



Touch



Smell



Light

Response

- A response is a reaction shown by an organism to changes occurring in its surroundings or inside the body.
- OR
- A response is the reaction of an organism to a stimulus.
- Nervous tissue plays an important role by using receptors to detect stimuli and effectors to produce appropriate responses efficiently.



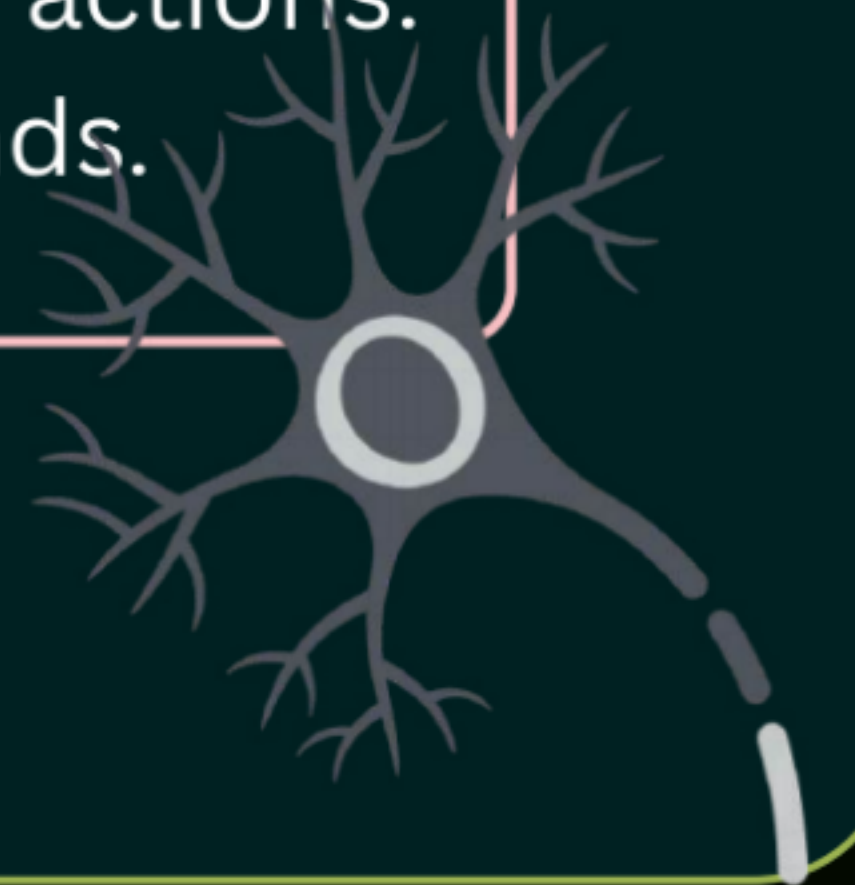
Receptors and Effectors

Receptors

Receptors are specialised cells or sense organs that detect stimuli from the environment and convert them into nerve impulses.

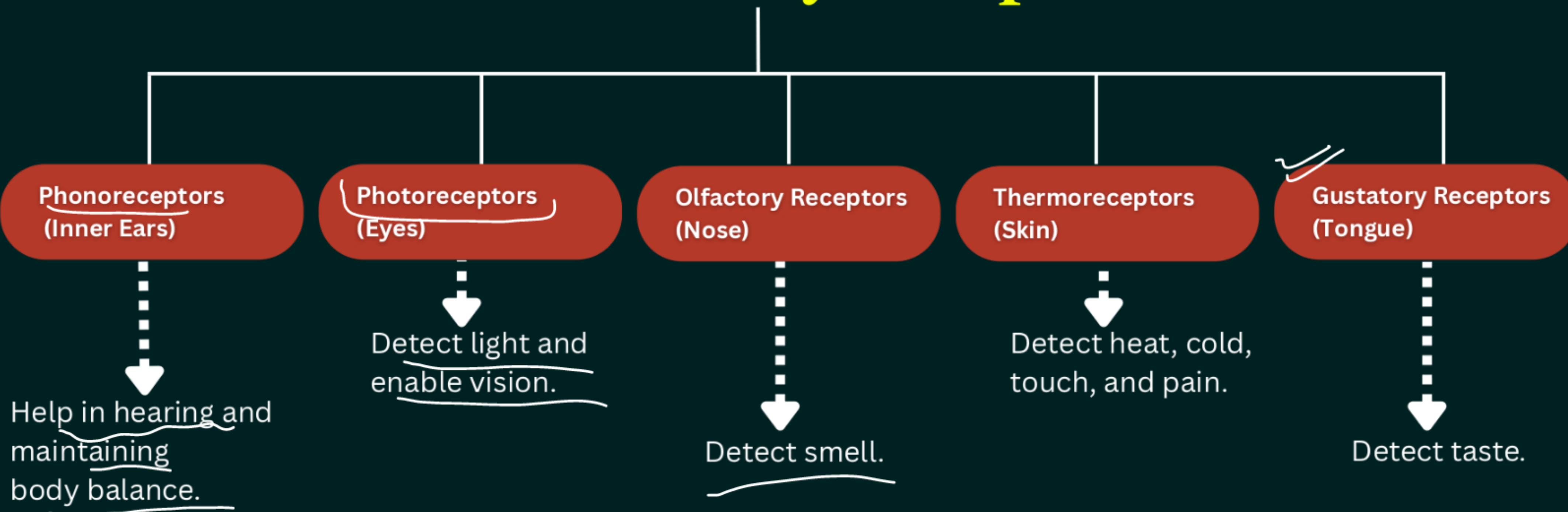
Effectors

Parts of the body that respond to impulses sent by the nervous system, converting them into actions. e.g. muscles and glands.



Receptors

5 Basic Sensory Receptors



Receptors and Effectors

Stimulus (e.g. touching a hot object)



Receptors (e.g. Thermoreceptors in skin detect heat)



Sensory Neuron (carries signal to spinal cord/brain)



Central Nervous System (Brain/Spinal cord processes the signal)



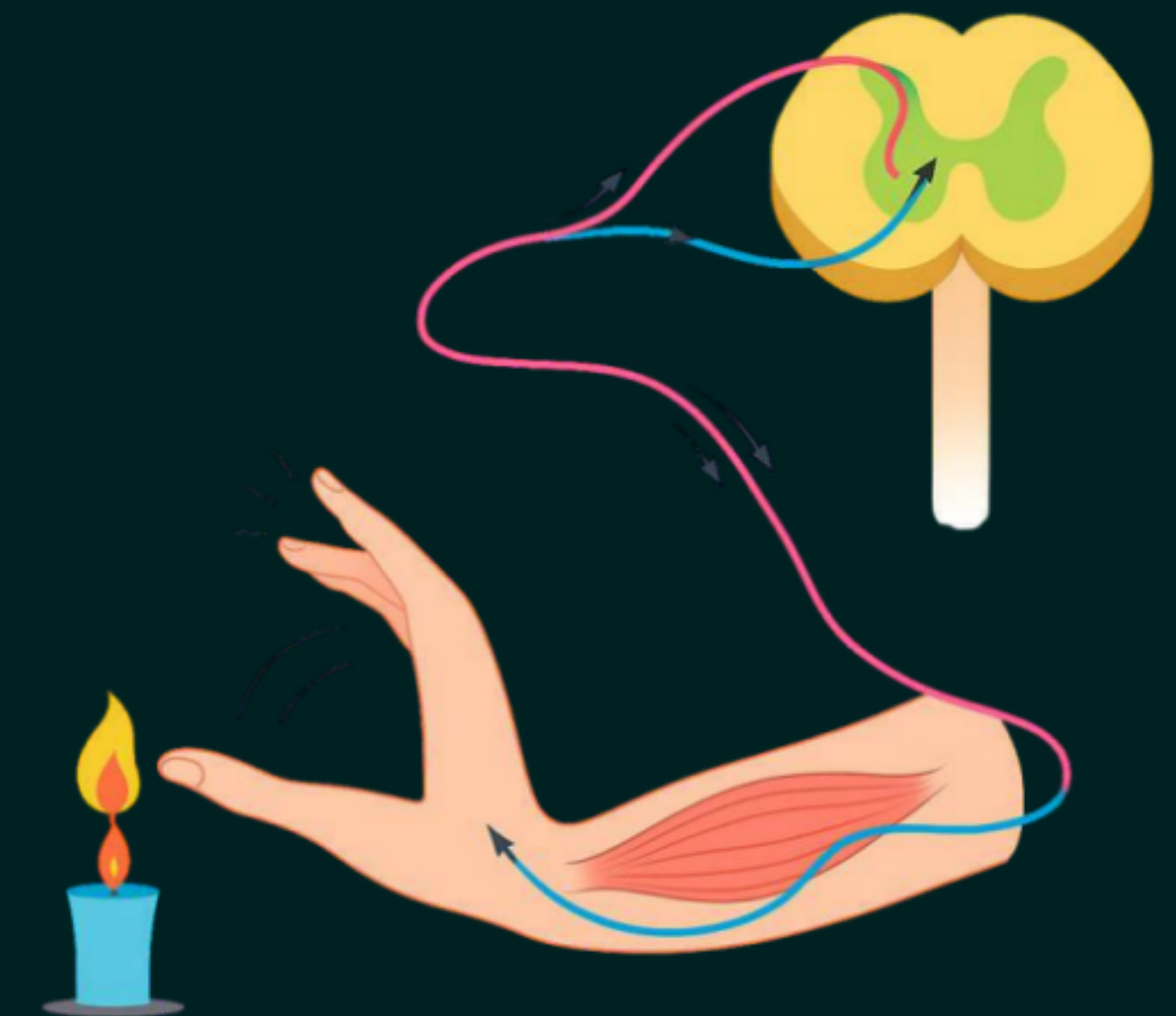
Motor Neuron (carries command from CNS to effector organ)



Effector (e.g. muscles in hand)



Response (e.g. withdrawing hand quickly)



Neuron

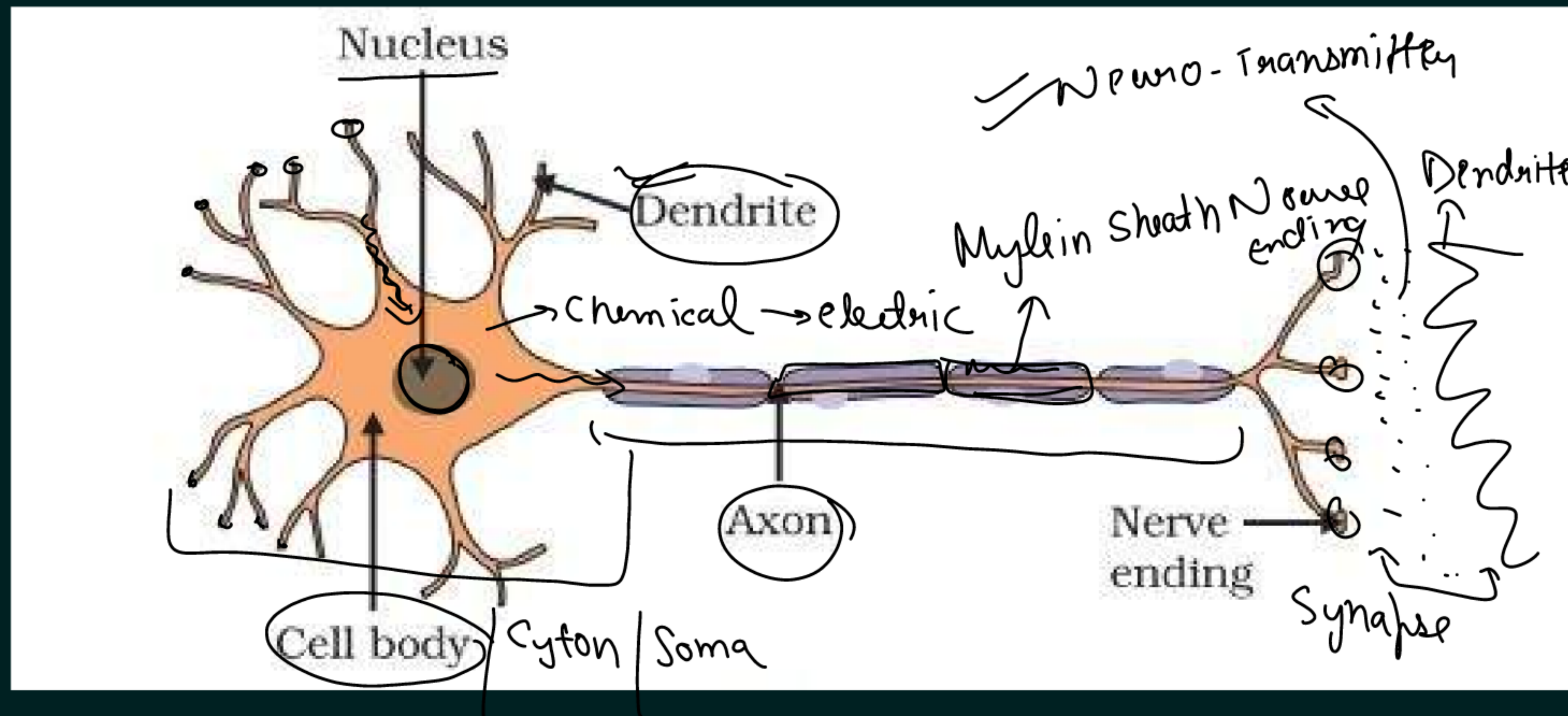
- **Neurons are the basic, structural and functional units of the nervous system.**
- They are specialized cells that transmit information in the form of electrical impulses.

Functions of Neurons:

- Receive signals from sensory organs (eyes, ears, skin, etc.).
- Process and interpret those signals.
- Send responses to effectors (muscles and glands) for action.

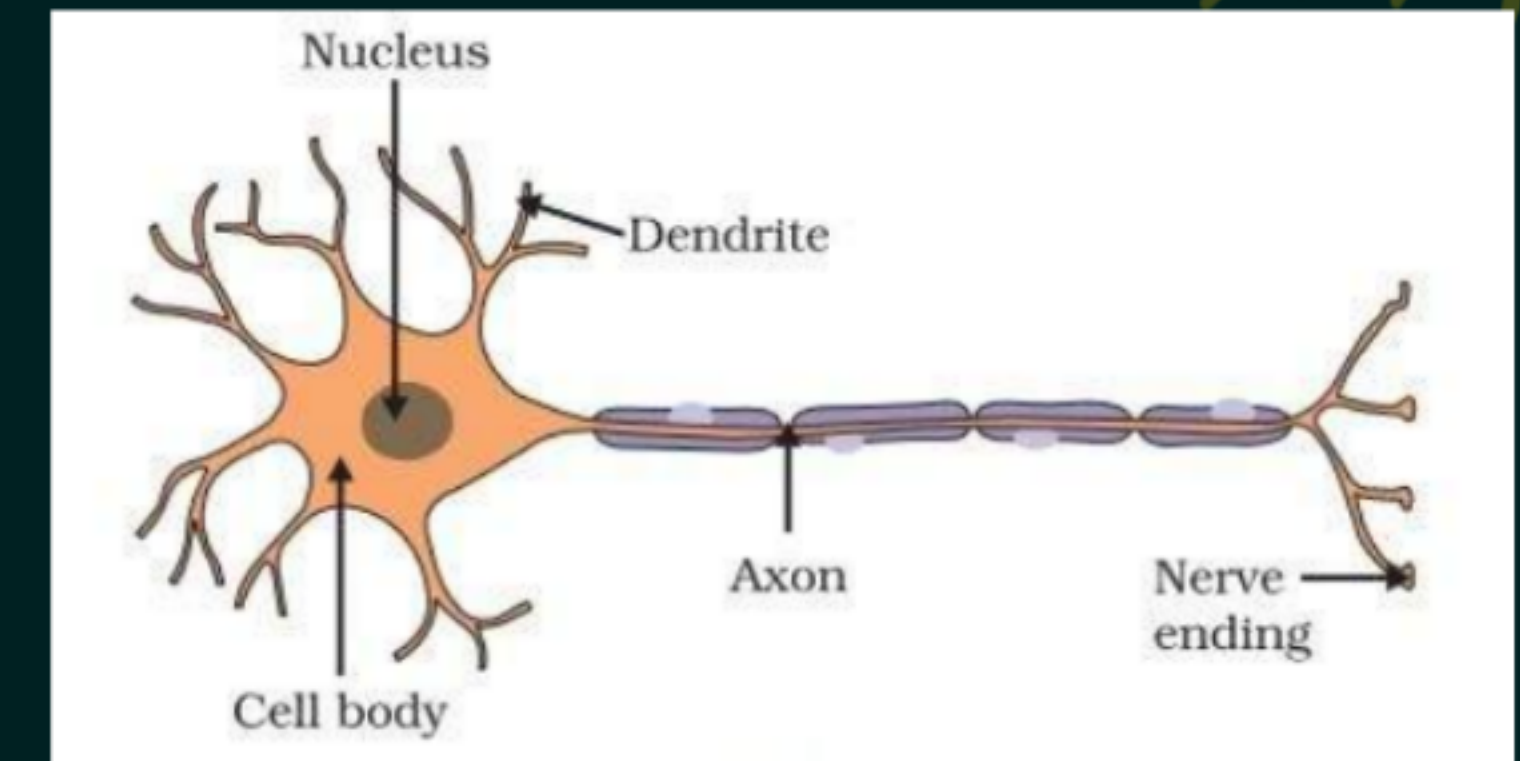


Parts of Neuron



Parts of Neuron

1. **Dendrites:** Branched extensions that receive signals from other neurons and pass them to the cell body.
2. **Cyton / Cell Body:** The main part of the neuron containing the nucleus and cell organelles such as mitochondria and Golgi bodies.
3. **Axon:** The longest fibre extending from the cell body of a neuron; it carries electrical impulses from the cell body to the axon terminal, which passes the impulse to the next neuron's dendrite.
4. **Myelin Sheath** - It is a fatty layer around the axon of some neurons.
 - It acts as an insulating layer and speeds up the transmission of nerve impulses.



Parts of Neuron

5. Nerve Endings

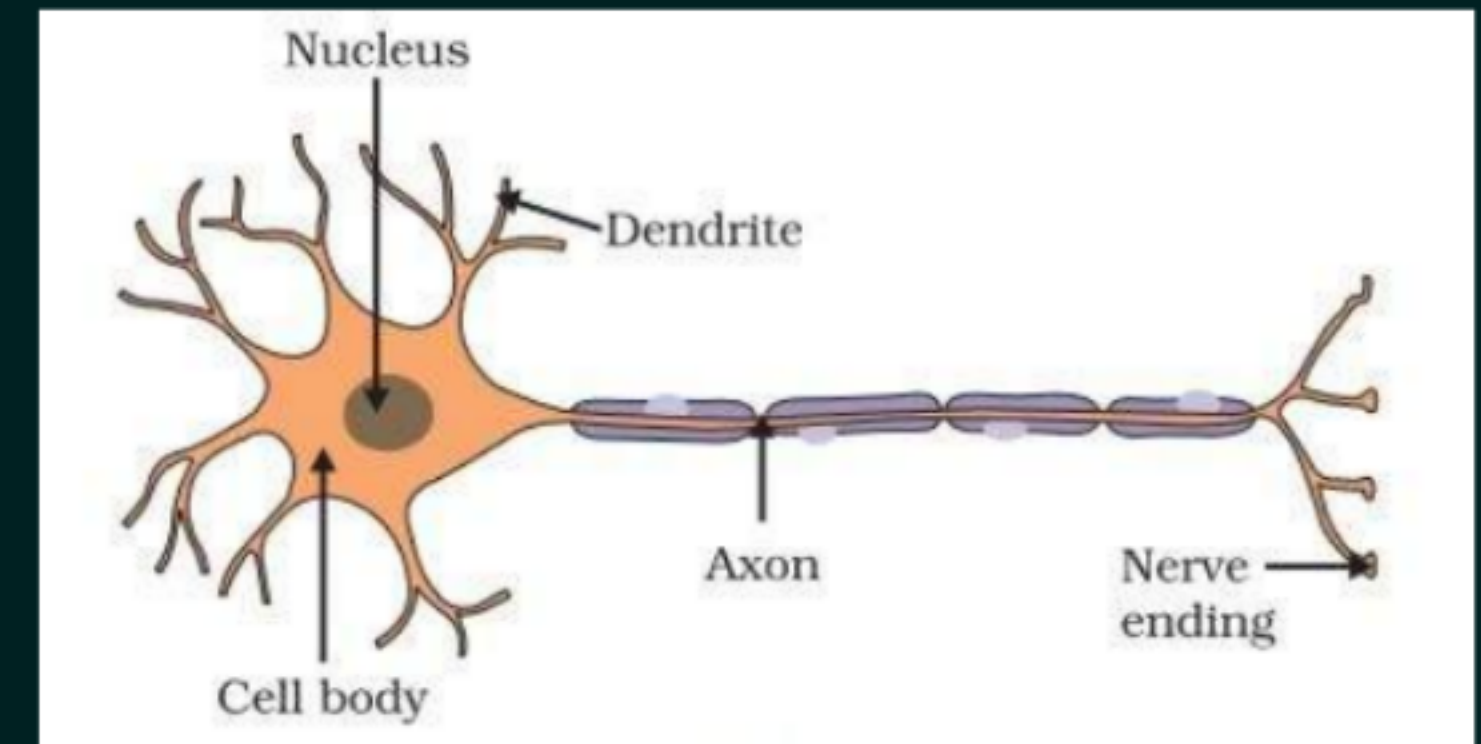
The terminal branches of an axon that connect with muscles, glands, or other neurons.

6. Synapse

A small gap between two neurons through which nerve signals are transmitted using chemicals called neurotransmitters.

7. Nerve Impulse

A nerve impulse is an electrochemical signal generated after stimulation that travels along a neuron to transmit information throughout the nervous system.



Passing of signals through and between neurons

Dendritic ends of the neuron receive the incoming information.



This triggers a chemical change, generating an electrical signal.



The electrical signal moves from the dendrite toward the cell body.



From the cell body, it travels through the axon.



The signal reaches the axon terminals (nerve endings).



At the terminal, it activates neurotransmitters to cross the synapse



These chemical messengers stimulate the next neuron's dendrite, continuing the process.





**Mann lga ke padhte
raho...milte hai agle
lecture me!**

